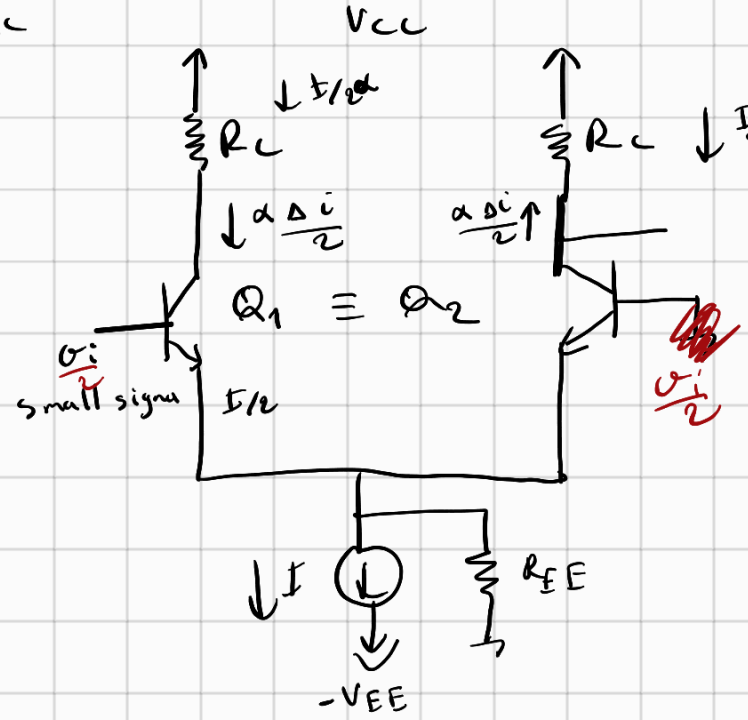
In ac

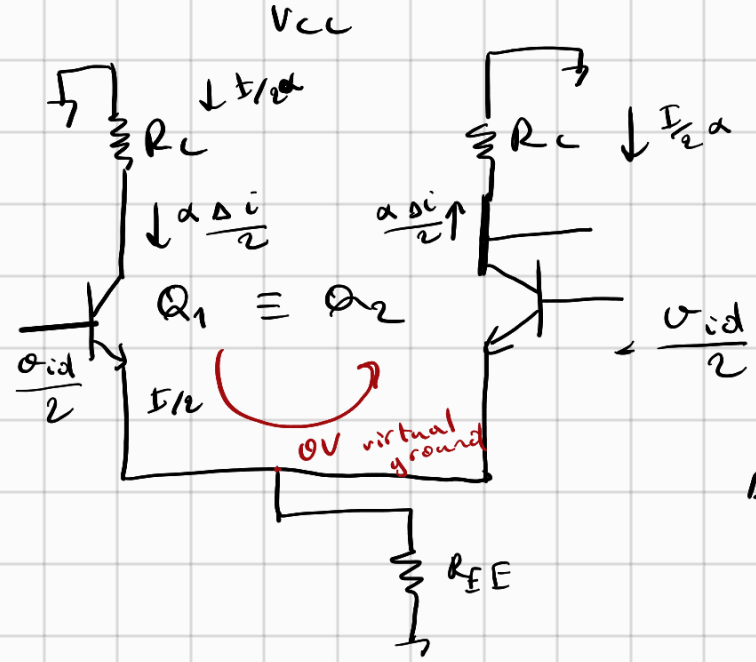


$$v_1 = V_{cc} - \frac{\alpha R_c I}{2} - \frac{\alpha \Delta i}{2}$$

$$v_2 = V_{cc} - \frac{\alpha R_c I}{2} + \frac{\alpha \Delta i}{2}$$

$$\boxed{v_o = v_2 - v_1 = \alpha \Delta i R_c}$$

$$\Delta i = g_m v_i$$



$$\Delta i = g_m V_{id}$$

$$v_2 = \frac{g_m v_{id} R_c}{2}$$

$$v_1 = -\frac{g_m v_{id} R_c}{2}$$

$$v_d = g_m v_{id} R_c = v_2 - v_1$$

$$\boxed{A_d = g_m R_c = \frac{v_d}{v_{id}}}$$

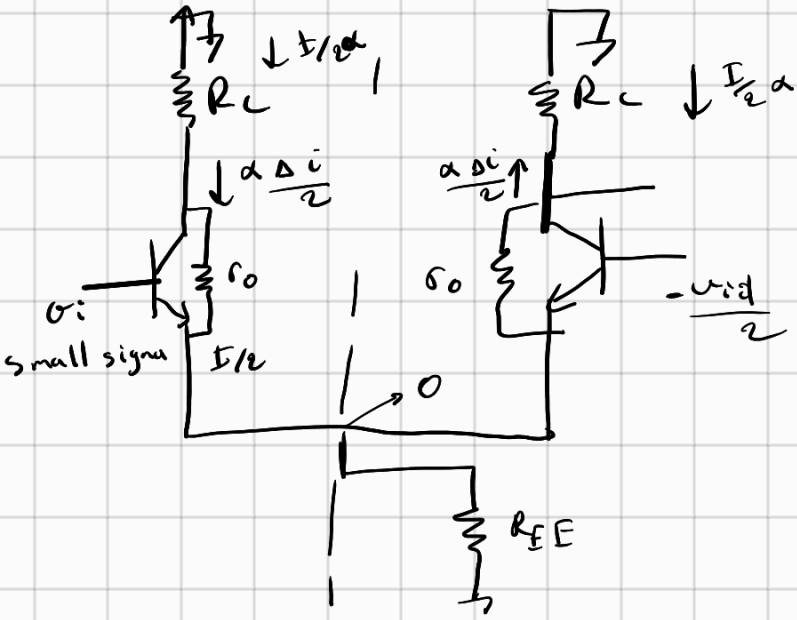
differential

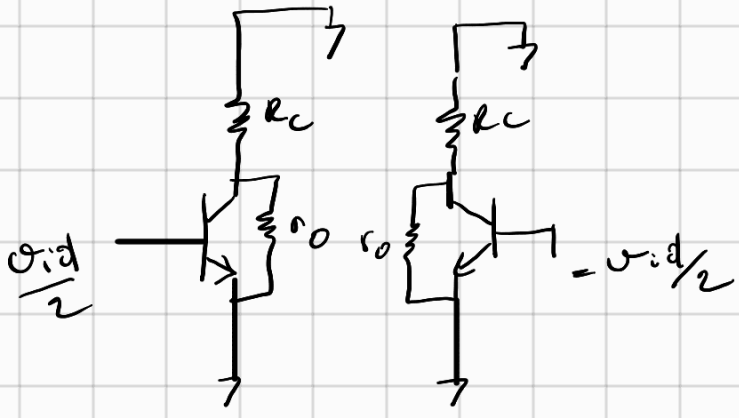
$$\boxed{g_m = \frac{I_c}{V_T}}$$

$A_d = \frac{I_C}{V_t} \quad R_C = \frac{I_C R_C}{V_t}$

soğukta kazanç artar

Effect of R_o?

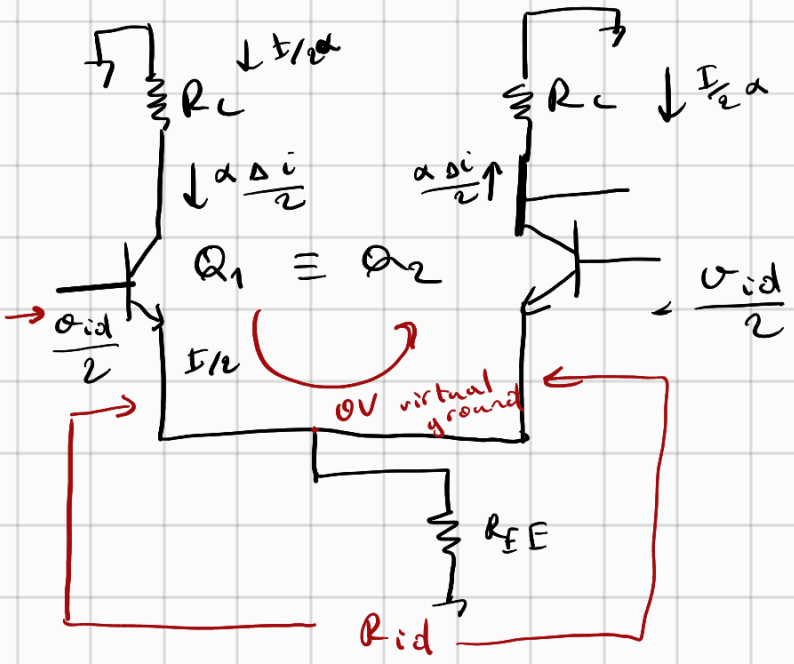


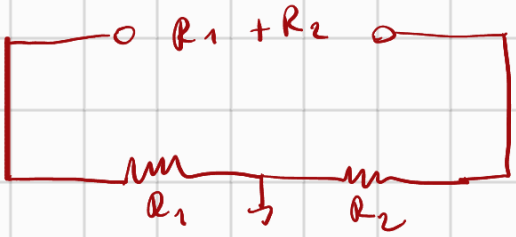


$v_1 = -\frac{v_{id}}{2} g_m (r_o \| R_C) \qquad v_2 = \frac{v_{id}}{2} g_m (r_o \| R_C)$

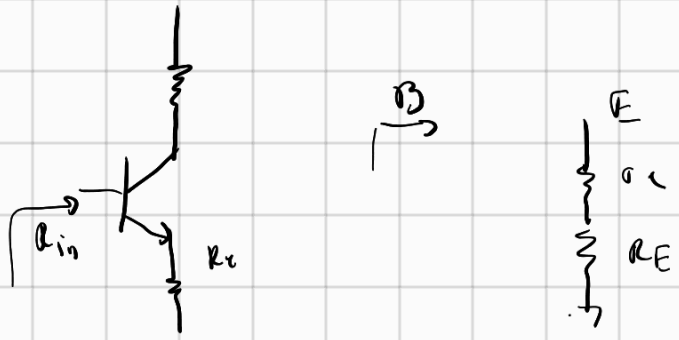
$v_d = g_m (R_C \| r_o) v_{id} \Rightarrow A_d = \frac{v_{ol}}{v_{id}} = g_m (R_C \| r_o)$

$\cong g_m R_C$

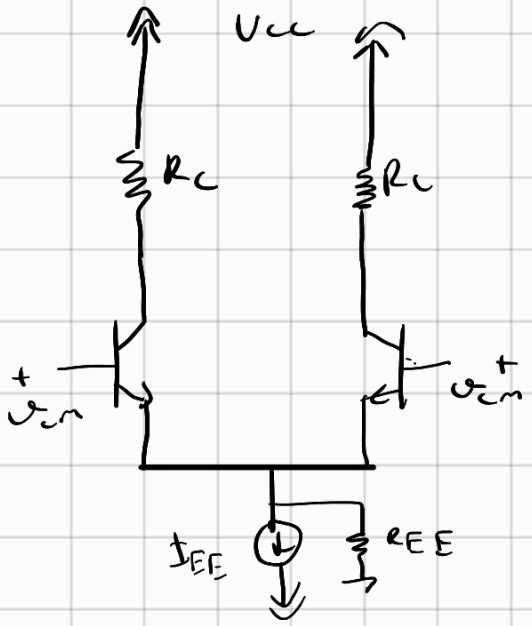




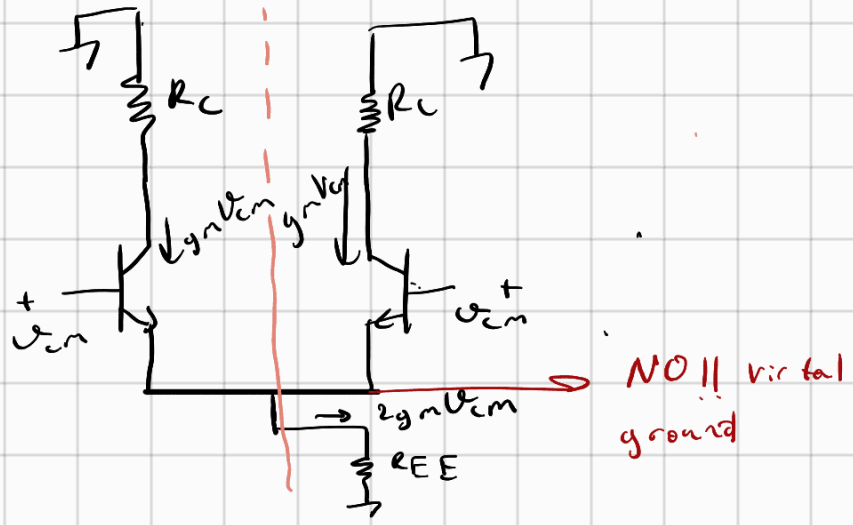
$$R_{id} = 2(\beta+1)(r_e + R_E)$$

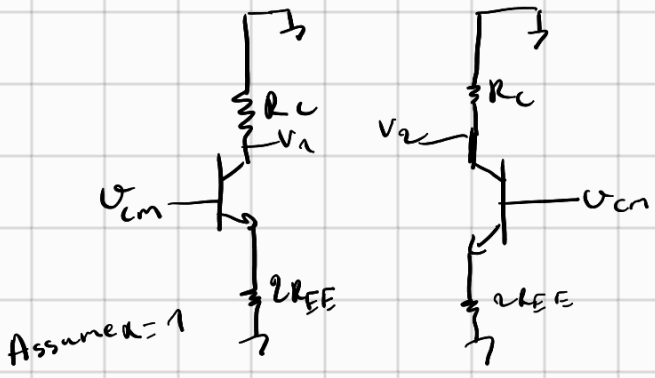


CMRR and Common mode operation



Ac mode





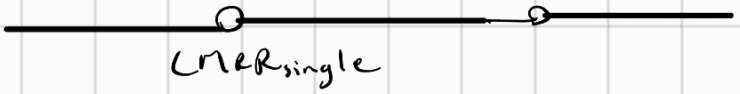
$$A_{cm\ single} \quad \frac{v_1}{v_{cm}} \simeq -\frac{R_c}{2R_{EE}+r_e} = -\frac{R_c}{2R_{EE}} \qquad \frac{v_2}{v_{cm}} \simeq \frac{-R_c}{2R_{EE}+r_e} \simeq -\frac{R_c}{2R_{EE}}$$

$$A_{cm\ diff} = \frac{v_1 - v_2}{v_c} = 0$$

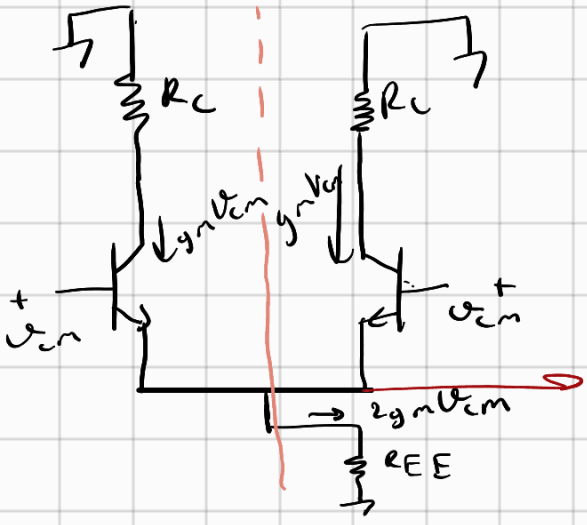
CMRR diff

$$CMRR = \frac{A_{diff}}{A_{cmdiff}}$$

$$CMRR = \frac{g_m R_c}{0} = \infty$$



$$\left|\frac{A_{d\,single}}{A_{cmsingle}}\right| = \frac{\left|\frac{1}{2} g_m R_c\right|}{\frac{R_c}{2R_{EE}}} = g_m R_{EE}$$

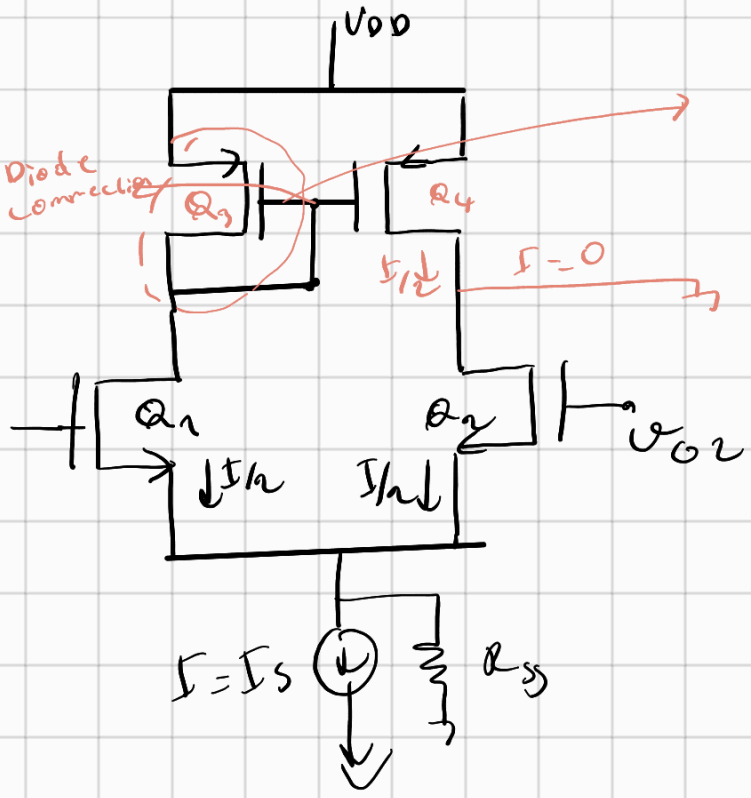


$$R_{icm} = ?$$

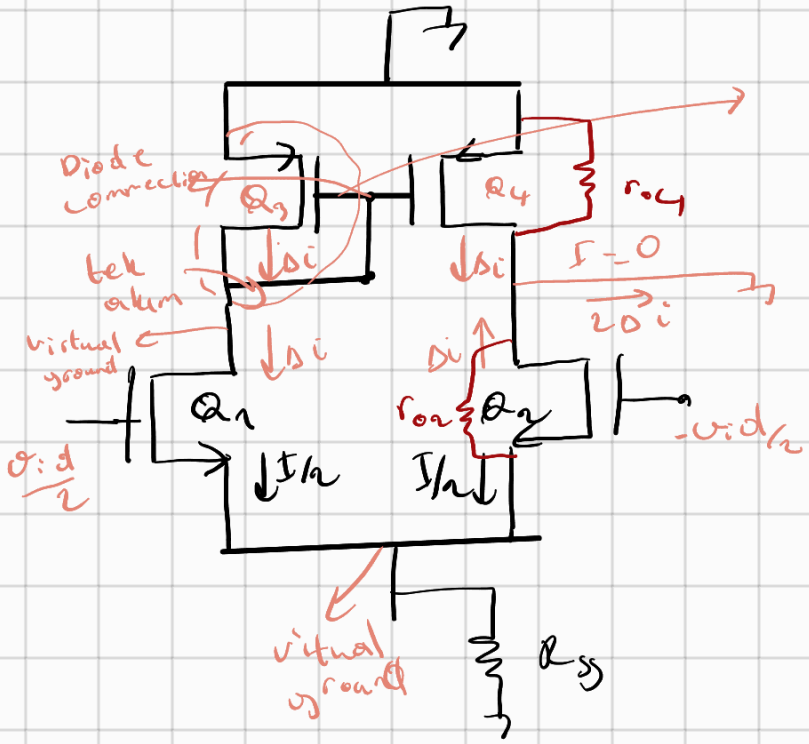
$$(\beta + 1)\left(R_{EE} + \frac{r_e}{2}\right)$$

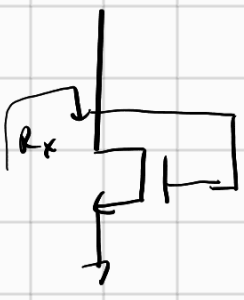
$$= (\beta + 1)(R_{EE})$$

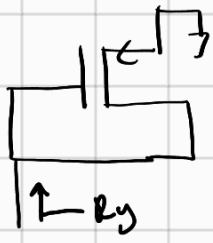
Active-Loaded CMOS differential Amplifier

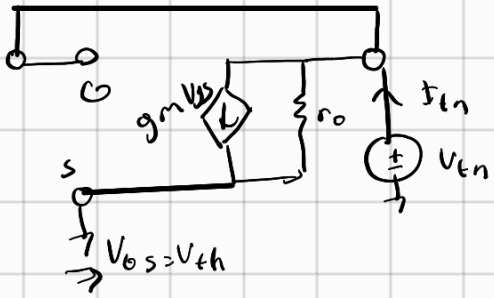










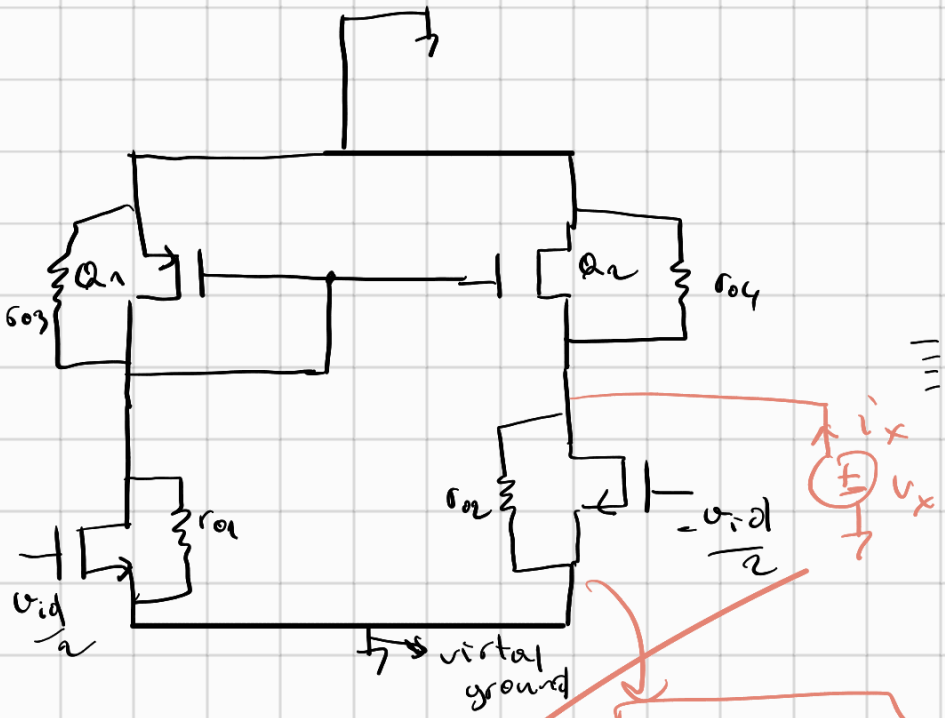


$$I_{th} = g_m v_{gs} + \frac{v_{gs}}{r_o}$$

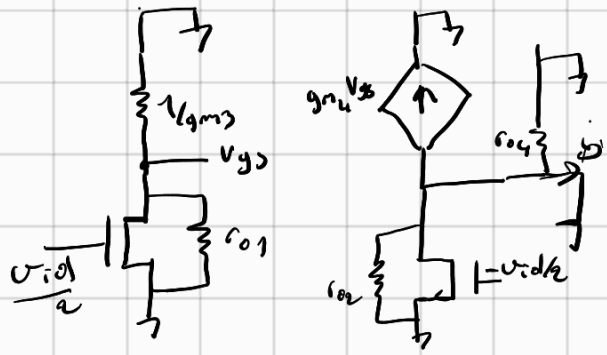
$$\frac{V_{th}}{I_{th}} = \frac{1}{g_m + 1/r_o}$$

(Single-ended)

Sadece sağ taraftan çıkış alabiliriz. Peki neden böyle bir şeye ihtiyaç duyuyoruz? Diğer devremiz sadece $= 1/g_m$ tek giriş alıyor. Onaillotiyoruz.



$\equiv$



$$v_{gs3} = -g_{m1}\frac{v_{id}}{2}\left(\frac{1}{g_{m3}} \| r_{o1}\right)$$

$$\simeq -\frac{g_{m1}}{g_{m3}}\frac{v_{id}}{2}$$

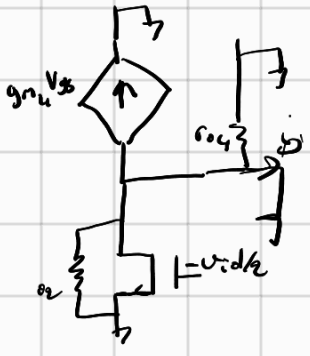
$G_m = g_m$

$$i_o = -g_{m4} v_{gs3} + g_{m2}\, v_{id}/2$$

$$i_o = \frac{g_{m4}\, g_{m1}}{g_{m3}}\frac{v_{id}}{2} + g_{m1}\frac{v_{id}}{2}$$

$$= g_{m1} v_{id}$$

$$g_{m1} = g_{m2},\ g_{m3} = g_{m4} \quad = g_{m1,2}\, v_{id}$$



$$\frac{v_x}{i_x} = R_o \simeq (r_{o2} \| r_{o4})$$

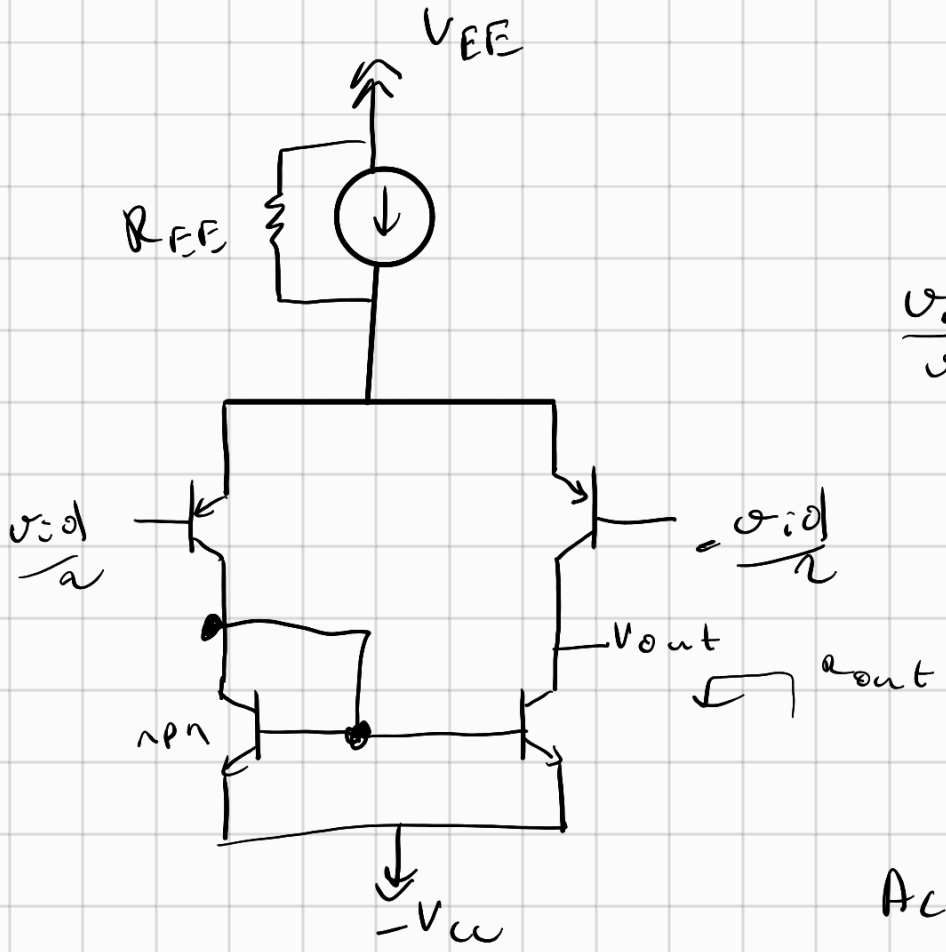
$$\frac{v_{out}}{v_{id}} = g_m (r_{o2} \| r_{o4})$$

$$r_{o2} = r_{o4} = r_o$$

$$\Rightarrow \quad = \tfrac{1}{2} g_m r_o$$

$$A_{cm} \simeq -\frac{1}{2 g_{m3} R_{SS}}$$

$$CMRR = (g_m r_o)\, g_m R_{SS}$$



$G_m = g_{m1} / g_{m2} = g_m$

$R_{out} = r_{o2} \| r_{o4}$

$\frac{v_{out}}{v_{id}} = g_m (r_{o2} \| r_{o4})$

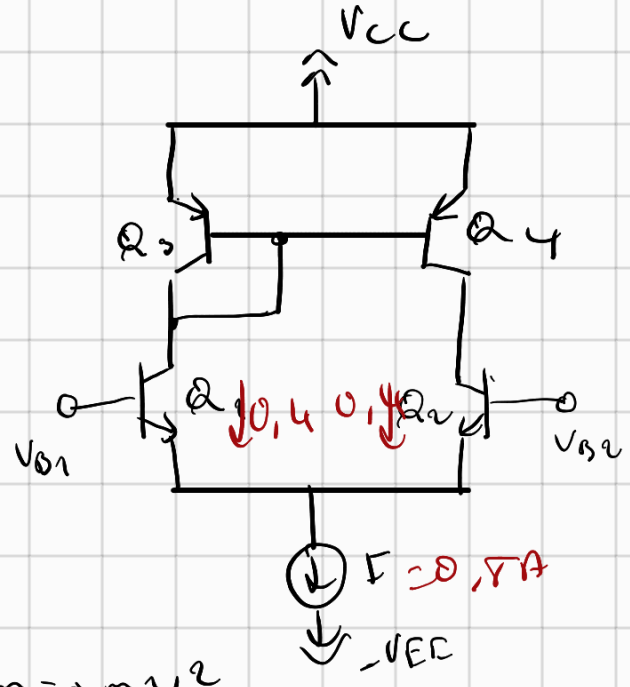
$r_{o2} = r_{o4} = r_o$

$\frac{v_{out}}{v_{id}} = \frac{1}{2} g_m r_o$

p type'ın dahası noise var

$A_{CM} = -\frac{r_{o4}}{\beta_3 R_{EE}}$

$CMRR = \frac{1}{2} \beta_3 g_m R_{EE}$



Consider BJT diff amp $I = 0,8\,mA$ $V_A = 100V$ $\beta = 160$. Find G_m, R_o, A_d and R_{id}. Find R_{EE}, A_{CM} and CMRR when the current source is implemented by a single npn current mirror

$\left[A_{CM} = -\frac{r_{o4}}{\beta_3 R_{EE}} \right]$

$G_m = g_{m1,2}$

$g_m = \frac{I_C}{V_T} = ?$ $V_T \to 25mV$

$g_{m1,2} = \frac{0,4mA}{25mV} = 16mS$

$$R_o = r_{o2} \,||\, r_{o4}$$

$$= \frac{V_A}{I_C} \,||\, \frac{V_A}{I_C} = \frac{1}{2} \cdot \frac{100}{0{,}4} = 125\,k\Omega$$

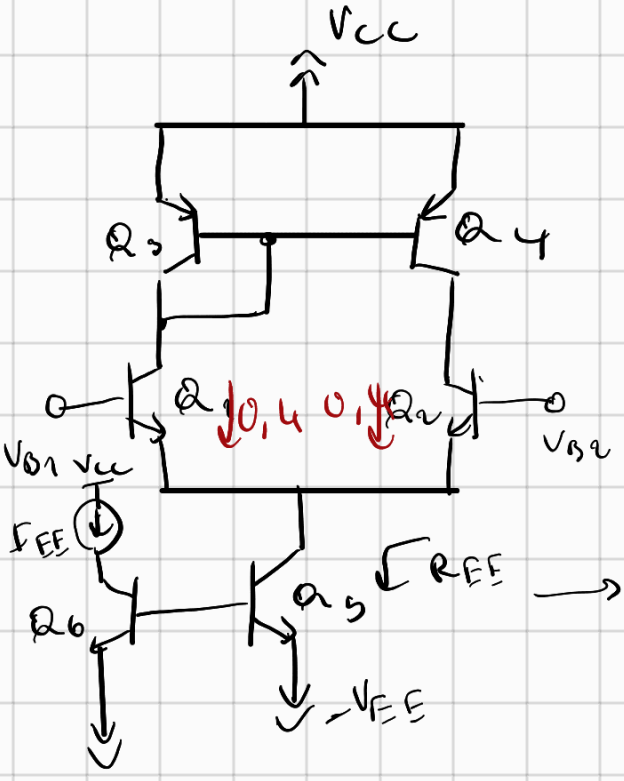
$$A_d = G_m R_o = 16\,m\mathcal{V} \cdot 125\,k\Omega$$

$$A_{dl} = 2000\,[V/V]$$

$$R_{id} = 2(\beta+1)(r_e) \qquad r_e = \frac{1}{g_m}$$

$$= 2 \times 160 \cdot 62{,}5$$

$$R_{id} = 20\,k\Omega$$



$$R_{EE} = r_{o5} = \frac{100}{0{,}8} = 125\,k\Omega$$

$$|A_{CM}| = \frac{250k}{160 \cdot 125k} = \frac{1}{80}$$

$$CMRR = \frac{A_d}{A_{cm}} = \frac{2000}{\frac{1}{80}} = 160\,000$$

$$= 20 \log 160\,000$$

$$= 104{,}0800$$